

Propositional Logic Series

Episode 3: Formal Definitions and Notation

What Makes an Argument Valid?

1. Logical Consequence and Entailment

Logical consequence / entailment

A conclusion is a *logical consequence* of some premises if and only if there is no possible case in which all those premises are true and the conclusion is not true. Equivalently, the premises *entail* the conclusion.

Entailment is a logical relation between premises and a conclusion. It is not physical causation. It is also different from inference: a reasoner *infers* a conclusion, while premises *entail* a conclusion.

2. Validity, Invalidity, and Counterexamples

Valid argument

An argument is valid if and only if there is no possible case in which all its premises are true and its conclusion is false.

Invalid argument

An argument is invalid if and only if there is at least one possible case in which all its premises are true and its conclusion is false.

Counterexample to an argument

A counterexample to an argument is a possible case in which all of the argument's premises are true and its conclusion is false.

One counterexample is sufficient to establish invalidity. To establish validity, every possible counterexample must be ruled out. Merely failing to find a counterexample after a quick search is not a proof of validity.

A preliminary use of possible case

In this episode, a possible case is treated as a hypothetical scenario. This is not yet the complete mathematical definition. Later episodes will use valuations to represent cases precisely.

3. Three Kinds of Validity

Nomologically valid argument

An argument is nomologically valid if and only if it has no counterexample that respects the laws of nature.

Conceptually valid argument

An argument is conceptually valid if and only if it has no counterexample that respects the relevant conceptual connections between expressions.

Formally valid argument

An argument is formally valid if and only if its validity depends on its logical form rather than on the specific subject matter of its expressions.

Propositional logic is primarily concerned with **formal validity**. It replaces specific subject matter with propositional letters while preserving logical structure.

4. Bivalence

Bivalence

In every case considered, every proposition under consideration is either true or false, and not both.

This series works, for now, within classical two-valued logic. Other logical systems may use different frameworks, but they are not surveyed in this episode.

5. Actual Truth and Validity

Actual truth asks	Validity asks
Are the individual premises and conclusion true in fact?	Can all the premises be true while the conclusion is false?

- A valid argument may have a false premise and a false conclusion.
- An invalid argument may contain premises and a conclusion that are all actually true.
- Therefore, the claim that all the statements are true does not establish validity, and the claim that the conclusion is false does not establish invalidity.

6. Sound Arguments

Sound argument

An argument is sound if and only if it is valid and all of its premises are actually true.

The conclusion of a sound argument must be true. This is a consequence of the two defining conditions, not a third independent condition.

Argument	Premises actually true?	Classification
Valid	Yes	Sound
Valid	No	Not sound
Invalid	Yes or no	Not sound

Two basic ways to challenge an argument are:

1. show that at least one premise is not true; or
2. show that the argument is invalid by giving a counterexample.

7. Deductive Does Not Mean Valid

A **deductive argument** presents its conclusion as something that cannot be false if all its premises are true. **Validity** tells us whether that claim succeeds. A deductive argument can therefore be invalid.

Unless stated otherwise, valid in this series means deductively valid. Inductive arguments are not presented as guaranteeing their conclusions and are evaluated by different standards.

8. Notation Used in This Episode

Notation	Use
P, Q, R	Propositional letters standing for complete propositions.
$\neg P$	Not P .
$P \vee Q$	P or Q , using inclusive or unless stated otherwise.
$P \rightarrow Q$	If P , then Q .
$\therefore Q$	Marks Q as the conclusion of the displayed argument. The symbol is not a connective.

Example of a valid form:

$$\frac{\begin{array}{c} P \rightarrow Q \\ P \end{array}}{\therefore Q}$$

Example of an invalid form:

$$\frac{\begin{array}{c} P \vee Q \\ P \end{array}}{\therefore Q}$$

The second form has a counterexample when P is true and Q is false. Named inference rules are not introduced in this episode.

9. Counterexample Search Checklist

1. Identify every premise and the conclusion.
2. Try to make the conclusion false.
3. At the same time, keep every premise true.
4. If such a case exists, stop: the argument is invalid.

Common error: Making only some premises true is not enough. A counterexample must make *all* premises true together and the conclusion false.

Sources and Note

The controlling conceptual and pedagogical source for this episode is P. D. Magnus, Tim Button, Robert Trueman, and Richard Zach, *forall x: Calgary: An Introduction to Formal Logic*, with contributions by J. Robert Loftis and Aaron Thomas-Bolduc, Chapter 2.

Official page: <https://forallx.openlogicproject.org/>

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